CHAPTER 16

DATA ANALYSIS

Introduction

Each day we come across different kinds of information. For example, this year production of rice increased by 8 percent; who was the best hockey player last year or, how many mobile phones were sold by company 'A' in the month of January? Similarly, from time to time we make use of printed information, for example, train timetables showing time of arrival and departure from stations, price of fruit-vegetables, current price of petrol, production amounts of grain and dairy, production of steel, coal etc. There are lots of other data which we use to plan for future and to take important decisions.

How to find different types of information

Can we tell today about the expected temperature of next two days? Or, the quantity of rice produced this year in our state? Or, what was the increase (or decrease) in price of petrol in the last five years?

It may not be possible to give the exact answer immediately but we can say something about these questions after analyzing relevant data.

We know that different types of data are published in newspapers and magazines, for example, data related to production of grains, weather-related information, details of games and sports, prices of food products etc.

Data related to health and education is collected by different departments and organizations under the government. In health, the government may have data on the areas most affected by a certain disease, the number of people suffering from a particular disease etc. On the basis of this information, we are able to decide the best measures to prevent the spread of a disease. Till about 6-7 decades ago, one of the most important questions for India was the quantity of grain produced every year, the quantity required to feed our population, and the quantity of grain that might have to be imported from other countries to meet our needs.

Try These

1. Besides mathematics, you must have come across data in other subjects like science, social studies etc. Give few examples of data in different subjects.
2. Look at some newspapers and magazines and collect examples of data printed in them. Discuss among yourselves the topics covered by these data.
3. Find which types of data are available in your school-office?
4. You might have seen data displayed on the notice board in your school premises. Which types of data you have seen?

Think and Discuss

Write the different sources from where you can get the answers of the following questions:

1. Which is the disease that most affects your district?
2. What is the population of your district in the current year?
3. In the current year, what is the minimum market rate of wheat and rice fixed by the government?

Some more questions

We also want to know the answers of many questions related to ourselves like, how fast can I run; can I run faster than my classmates? Am I taller (or shorter) than the other students in my class? How can we find the answer of questions like these? There may be a few students who run faster than you and a few who run slower; some will be taller and some shorter.

Rani's height is 160 cm. The height of her classmates, in cm, is shown in the table given below:

161	160	162	159	161	158	162	163
158	158	160	159	160	161	163	160
158	161	158	159	163	159	160	159
158	160	159	162	163	160	159	159
159	162	161	163	159	161	161	160
163	160	163	161	160	158	160	163
160	160						

From the given table, can you tell where does Rani stand as compared to her classmates? It will be difficult to compare her height with all students in her class. It will be easier to compare if we organise the given data. To organize our data, we will prepare a frequency table. Then our data will look as shown in table-1:

Table-1

Height (In *cm*)	158	159	160	161	162	163
Number of Students	7	10	13	8	4	8

What you interpret or conclude by looking at this table?

First thing that we can conclude is that maximum students are in the 160 *cm* group. This group contains 13 children. 17 children are in those groups whose height is less than Rani's height. Minimum height is 158 *cm* and there are 7 children in this group.

What else can we conclude from this table? Discuss with your friends and find at least 5 more conclusions.

In the same way if we talked who runs the fastest, then we find that the speed of all runners is not same. The table below shows the speed of 50 persons in km per hour i.e. it describes the number of kilometer covered by each runner in one hour.

Table-2

Speed of running (km per hour)	15	11	9	5	6	4
Number of students	5	6	7	8	9	10

This means that if Nafisa can run at 7 km/h then we can do a comparative analysis of her speed with the help of the given table. We can also see that how many persons run faster than her and how many students run slower than her.

Try These

Read the questions given below. What kind of data do we need to find the answer of these questions? Discuss and write from where and how we will get these data?

1. How did the price of petrol vary during the last three years?
2. Which state in our country had minimum rainfall this year?
3. In what quantity production of fish increased in the last five years?
4. Population of which state was maximum in 2011?
5. What changes happen in last five years in your village and city?
6. Which district of Chhattisgarh has the maximum number of schools?
7. How many international hockey matches were played by India in the last five years?
8. How much rice was produced all over India from 2010 to 2015?

Think and discuss

1. If the number of students in your school is 1000 and you want to compare your height will all of them, how you will do it?
2. What kind of data will you need if you want to compare your running speed with all the students in your district?

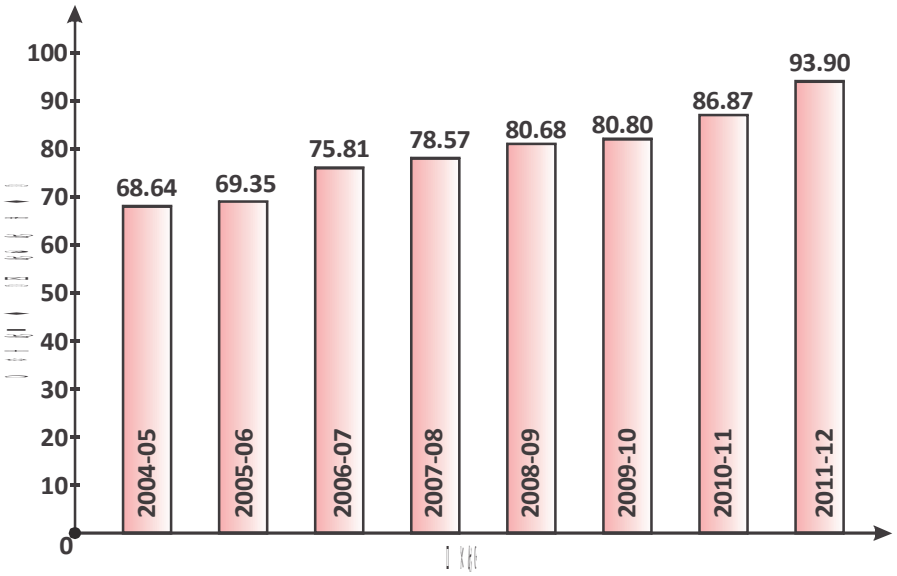
Graphical Representation of Data

You learnt organisation and representation of data in your previous class. You studied about frequency tables and graphical representation of data, in which we tried to learn how to make histograms, frequency polygons and cumulative frequency curves. We get many types of information and can draw several conclusions on the basis of these data representation. Let us consider more examples of these types of data in the following section.

Bar Graphs

The bar graph given below shows the production of wheat in a state in different years-

Which information can be seen clearly on considering this graph? Find out the answers of the following questions by reading the graph.



(i) How many tons of wheat was produced in 2007-08?

(ii) In which year was the production of wheat maximum?

(iii) Can we say that the production of wheat increased every year?

(iv) The production of wheat showed greatest change in which two years?

Try These

Reading a table

The table given below shows the amount of annual rainfall in a particular city.

Year	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
Total Rainfall	24-7	21-2	14-5	13-2	12-1	16-8	19-9	29-2	31-6	21-0

(in inches)

Study these figures related to rainfall, draw their bar graph and find at least five conclusions based on the graph.

Mean, Median, Mode

You found the answer to some questions and drew conclusions from bar graphs and frequency curves. Can you tell that what was the average annual production of wheat during 2005 to 2015 by looking at the bar graph? Can you tell which year will fall in the middle if we arrange the data in order? From the second table we can find the average rainfall in a city in a year or generally how much annual rainfall is expected. We cannot say anything about the order and trend of data only by looking at bar graphs. To find average wheat production we need arithmetic mean. Recall, how we can calculate mean.

Arithmetic Mean

Now, to find average production of wheat from the year 2005 to 2012 we add the wheat production (in lacs ton) of these years then divide it by the number of years. Let us see what is the average value of annual wheat production.

Total production = 68.64 + 69.35 + 75.81 + 78.57 + 80.80 +80.80 + 86.87 + 93.90

= 634.74 lacs ton

Number of years from 2005 to 2012 = 8 years

$$\text{Average production} = \frac{634.74}{8}$$

= 79.34 lacs ton

Here, we calculated the average value of wheat production. In statistics, average of data is known as arithmetic mean. So if we want to find arithmetic mean then we add the data and divide the sum by the number of data points. The following formula is used to show this relation:

$$\text{Arithmetic Mean} = \frac{Sum\ of\ data}{Total\ number\ of\ data}$$

If data is written as x, then sum of data is Σx and number of data is n. Then

$$\text{Arithmetic mean} = \frac{\Sigma x}{n}$$

Arithmetic mean is generally denoted by A.M., M and $\overline{X}$.

Discrete Series

Data considered till now was individual points in a series and the number of data-points was small. If the number of data-points is larger then how can we calculate arithmetic mean?

Marks obtained in mathematics by 35 students from Class-IX are as follows:

30, 30, 38, 40, 42, 35, 40, 30, 45, 48,
40, 42, 38, 30, 38, 40, 35, 30, 42, 40,
42, 38, 35, 42, 40, 38, 42, 40, 48, 45,
38, 40, 30, 35, 35

Here, minimum marks obtained is 30 and maximum is 48. It is seen that the marks are limited to 35, 35, 38, 40, 42, 45, 48 and they are repeated again and again. Therefore, these figures can be written in the following way:

Marks Obtained (x) :	30	35	38	40	42	45	48
Frequency (f) :	6	5	6	8	6	2	2

When we have this type of data then for calculating arithmetic mean we multiply the data-points (observations) with their corresponding frequency and add them. This sum is divided by the sum of frequencies to get arithmetic means.

Marks obtained (x)	**Frequency(f)**	**Product of marks obtained and their corresponding frequencies**
30	6	180
35	5	175
38	6	228
40	8	320
42	6	252
45	2	90
48	2	96
	$\sum f = 35$	$\sum fx = 1341$

$\therefore$ Arithmetic Mean =

$$\frac{\textit{Sum of the product of marks obtained and corresponding frequency}}{\textit{Sum of frequencies.}}$$

$$\bar{X} = \frac{f_1x_1 + f_2x_2 + f_3x_3 + f_4x_4 + f_5x_5 + f_6x_6 + f_7x_7}{f_1 + f_2 + f_3 + f_4 + f_5 + f_6 + f_7}$$

$$= \frac{\sum f_i x_i}{\sum f_i}$$

$$= \frac{1341}{35} \qquad \bar{X} = 38.31$$

Average marks in mathematics of Class-IX student is 38.31.

Now we will discuss about arithmetic mean of data where observation points are repeated and also the number of observation points is very large. In this case, we distribute the data into groups to calculate arithmetic mean.

Let us consider an example.

Example-1: There are 100 students in the village middle school. The distance in *km* between their houses and school is given below. Find the average distance between home and school using this data.

17	1	19	0	4	1	3	2	0	4
5	7	2	8	9	19	2	17	1	18
0	3	2	5	2	8	1	10	1	11
13	8	9	4	15	0	15	3	11	11
2	19	0	14	12	1	12	1	13	1
9	3	6	4	14	3	10	12	4	8
0	7	9	6	5	9	7	8	2	9
5	8	6	7	9	5	5	6	3	8
7	5	0	1	3	0	4	2	0	1
3	0	4	3	2	0	1	0	4	0

Solution: If we look at the data, we find that many observations are repeated and the minimum value is 0 and the maximum value is 19. We will have to divide the data into group, so that calculations become easy.

Let us divide the data into groups at equal intervals, 4 in this case. We find that number of students whose homes are at distance between 0-4 *km* is 39, number of students who comes from 4 to 8 *km* is 24 and so on. Find the number of students in the group 8-12 *km*, 12-16, and 16-20.

Distance of school from the home of student (*km*)	Number of students (f)	Midpoint (x)	fx
0-4	42	2	84
4-8	24	6	144
8-12	19	10	190
12-16	9	14	126
16-20	6	18	108
	$\sum f_i = 100$		$\sum f_i x_i = 652$

In the table above, we calculated the midpoint of each interval by adding the upper and lower limits of each interval and dividing by 2. Now we add the product of midpoint and number of students to get the average.

$$\text{Average} = \frac{\textit{Sum of products of number of students and midpoint}}{\textit{Total Number of students}}$$

$$= \frac{84+144+190+126+108}{100}$$

$$= \frac{652}{100}$$

$$= 6.52\ km$$

We can consider average as a number that represents a single feature of the entire group of observations. Obviously, average is greater than the minimum data value and smaller than the maximum value and it lies somewhere in the middle of the whole data. This is 'arithmetic mean'.

Calculation of arithmetic mean

Let us consider some more examples to understand arithmetic mean.

Data related to production of pulses in five years is given in the table below:

Years	2007–08	2008–09	2009–10	2010–11	2011–12
Production of pulses (in lacs ton)	14.8	14.6	14.7	18.2	17.2

Here, we have to find the arithmetic mean or average. To find this we have to add all observation points and after that divide the sum by the total number of years.

$$\text{Arithmetic mean} = \frac{14.8+14.6+14.7+18.2+17.2}{5} \text{ lac tons}$$

$$= \frac{79.5}{5} = 15.9 \text{ lac tons}$$

Average production of pulses is 15.9 lac tons. Actual value of pulses produced in each year is different from the average annual production. By using average, we can give one value to show the production of pulses of five years.

Let us consider one more example of average.

Eample-2: The data for rainfall (in mm) in Dhamtari district are given. Find the average of this data.

880.5, 1474.9, 806.3, 1554.9, 1019,2, 1046.5, 1017.2

Solution : You know that

$$\text{Mean} = \frac{\textit{Sum of observation – points}}{\textit{Number of observation – points}}$$

$$\text{So, average} = \frac{880.5+1474.9+806.9+1554.9+1019.2+1046.5+1017.2}{7}$$

$$= \frac{7799.5}{7} = 1114.21 \ mm$$

So, arithmetic mean of rainfall is 1114.21 *mm*

Uses of Average in day-to-day life

Can you tell how much time do girls generally get to play at home? We know that daily playing time is not fixed; some days one can play for many hours and on some days we may get only a few hours or no time at all to play.

This means that if we only observe one girl for one day then we can't say how many hours are spent on play by all girls. If you collect the data of playing time over several days for several girls then you will have large quantities of data. It will not be easy to organise this data. To solve the given problem, we can consider the data over one month and find average daily playing time. Consider table-3. Here we are given the time spent playing of 50 girls. Can you tell how much time is spent per day by most girls on sports and games?

In table-3 you can see that on average, most girls get less than 2 hours to play. The maximum number of girls, 12, on an average play for at least 2 hours per day but the average daily playing time is not 2 hours for all the girls. To find the arithmetic mean in the given case, we first need to know the average daily playing time of all 50 girls individually.

Table-3

Average daily play time (in hours)	Number of girls	Total play time for 50 girls
x_i	f_i	$x_i f_i$
0	4	0
$\frac{1}{2}$	6	3
1	8	8
$1\frac{1}{2}$	9	$13\frac{1}{2}$
2	12	24
$2\frac{1}{2}$	7	$17\frac{1}{2}$
3	4	12
कुल	$\sum f_i = 50$	$\sum f_i x_i = 75$

$$\text{Average} = \frac{\sum f_i x_i}{\sum f_i}$$

$$\text{Average} = \frac{75}{50} \text{ hours}$$

= 1 hour 50 minutes.

If the frequency of observations $x_1, x_2, x_3, \ldots . x_n$ are $f_1, f_{2,} f_{3, \ldots .} f_{n,}$ respectively then it means that observation x_1 occurs *fi* times, data x_2 comes f_2 time and so on.

For example, in this problem the number of girls who on an average play for 0 hours is 4 and the number of girls who on an average play ½ hours is 6 and thus $x_1 = 0$, $fi = 4$ and $x_2 = ½$, $f_2 = 6$

Now, sum of products of observations and their respective frequency

$\Sigma(x_i f_i) = f_1 x_1 + f_2 x_2 + f_3 x_3 + ... + f_n x_n$ and

Total number of observations = $f_1 + f_2 + ... f_n$

To find the mean we will divide the sum of products by the sum of frequencies. In this way, the mean is:

$$\text{Mean (Arithmetic mean)} = \frac{\sum f_i x_i}{\sum f_i}$$ where the value of i will from 1 to n

Sum is represented by the Greek letter “, called sigma. This represents sum therefore sum of frequency is represented by “f_i and sum of products of observations and frequency is represented by “ $x_i f_i$.

It means that on an average every girl gets 1 hour 50 minutes to play each day. Now let compare this average with the rest of the data.

Can you tell how many girls play more than the average time and how many girls play less than the average time? You can see that 27 girls play more than the average time and 23 girls play less than the average time.

We find that, similar to this example, when we need to study data on a large scale then averages help us to organise the data; like in the examples of average play-time of girls or the average running speed of different persons.

Try These

1. Use the data given in previous example to find the average height of a student in a class of 50. Compare this height with your own height.

2. Use the data of 50 students given in table-2 to find the average speed. What conclusions we can draw from it?

In the previous example, you learnt how to find the average of given observations but suppose you are given the average then can you find the unknown observations?

Let consider the example given below:-

Example-3. Average of following observations - 25, 39, 35, *f*, 46 - is 36. Find the unknown data point, *f*.

Solution: You know that

$$\text{Average} = \frac{25+39+35+f+46}{5}$$

$$\text{Average} = \frac{145+f}{5}$$

On putting value of average in the equation

$$36 = \frac{145+f}{5}$$

$$36 \times 5 = 145 + f$$

$$180 = 145 + f$$

$$180 - 145 = f$$

$$35 = f$$

Therefore, value of *f* is 35. Thus the data points are 25, 39, 35, 35, 46.

Example-4. Find out the average height of students with the help of data given in the table.

Height (in *cm*)	158	159	160	161	162	163
Number of students	7	10	13	8	4	5

Solution : We know that mean (Arithmetic mean)

Height (in *cm*) (*xi*)	Number of students (*fi*)	$(f_i x_i)$
158	7	1106
159	10	1590
160	13	2080
161	8	1288
162	4	648
163	5	815
	$\sum f_i = 47$	$\sum f_i x_i = 7527$

So arithmetic mean $= \frac{7527}{47}$

$= 160.15\ cm$

So average height of student is 160.15 *cm*

Try These

1. Find the average of first 15 natural numbers.
2. Price of petrol (in Rupees) in Bhubaneswar (Orissa) during March-April, 2010 is given below. Find the mean of the data.

   61.28, 62.08, 59.35, 56.28, 59.28
3. Data (in lac tons) regarding rice production of a state over an 8 year period is given. Find the average of given data.

   84.98, 93.34, 71.82, 88.53, 83.13, 91.79, 93.36, 96.69

What Averages Tell Us

We saw that average gives us some basic information that represents a property of the entire data. But does arithmetic mean give us the complete picture of the whole data?

Read the following statements:-

1. This year, the average day-time temperature was 23°C in the month of February.
2. Average cost of petrol over the last five years was Rs. 65.70.
3. Average age of Class-X students is about 15 years.

You must have read or heard many more statements of this type. When we hear about average temperature of a day or month; or average price of petrol then we get some general idea and can draw some conclusions. But there are several pieces of information which we can't find from average.

For example, in Statement-1, in the month of February, the temperature would have been more than 23°C some times and other times less than 23°C. Average does not tell what the temperature was on a particular day; or what was the maximum temperature during the month or what was the minimum temperature; was there a lot of variation in day-to-day temperature or was it more or less constant?

In Statement-2 too, petrol prices must have varied from time to time. Average price of petrol could not have been 65.70 each year. From average, we can't say anything about the price of petrol today? But still, if we know that the price of petrol does not fluctuate daily or frequently then from average we are able to estimate that price of petrol per liter would have probably varied from about Rs. 64 to Rs. 66.

From statement-3, we know that age of some students will be less than 15 years and more than 15 years for others; we can't get any more information from this statement.

Let us consider one more example of average:

Example-5: The salaries of seven workers are given below:

1400 1500 8400 8700 9000 9200 9400

Let us calculate the average salary.

You know that, Average $= \frac{1400+1500+8400+8700+9000+9200+9400}{7}$

$= \frac{47600}{7} = 6800$ Rupees

Average salary as per data is Rs. 6800.

But, is average able to appropriately represent the center of the data? None of the seven data-points is approaching or close to the average. From this average, we can find out how money is spent each month on paying salaries but can't say how much a particular employee gets.

We conclude that average is not helpful in understanding distribution of data.

Median

When the values of data-points are very different from each other, then we are not able to draw many important conclusions from mean. In such cases, we will use a new numerical representative called median. Median is the data-point that comes exactly in the middle of organised data.

Let try to understand median with an example and then consider its utility.

Consider salary data given in Example-5 -

1400, 1500, 8400, 8700, 9000, 9200, 9400

What is the median of this data? There are 7 terms in data series which means that the fourth term is the middle term. Therefore median of this data is 8700. Middle term of the data given us median; sometimes, median is better able to represent data in appropriate manner because median value is not affected by extreme values in the data.

Try This

Find the median of the following data:

1. 25, 21, 23, 18, 20, 23, 24

2. 113, 102, 95, 85, 110, 109, 106, 110, 115

Let understand some more important uses of median.

Example-6. 21 people appear for interview for 10 posts in an office. They get following marks out of 50 in the interview:-

25, 23, 45, 40, 42, 38, 32, 43, 47, 36, 28, 37, 35, 34, 42, 21, 27, 18, 39, 41, 40

How will we select the 10 persons for the job? What should we do?

We know that those 10 will be selected from 21 people who get maximum marks. To make this process easy we arrange data in ascending order which will as follows:-

18, 21, 23, 25, 27, 28, 32, 34, 35, 36, 37, 38, 39, 40, 40, 41, 42, 42, 43, 45, 47

In this data, the 11th term i.e. 37 is median which lies between 10 people from the beginning and 10 people from the end. So, for job those who get more than 37 marks will be selected where 37 is median of data.

Thus, if total number of data is n

Then, median of data will be $\left(\frac{n+1}{2}\right)^{th}$ term.

You can see that in examples 4 and 5, the total number of data-points is odd. If the number of data-points is odd we can use the above formula to easily find the median. But, if the total number of data-points is even then how can we get median? Let us try to understand this by an example.

Example-7. Height (in *cm*) of 10 students are following:-

117, 106, 123, 110, 125, 112, 115, 102, 100, 115

Find the median of this data.

Solution: To find the median, first arrange the data-points in ascending order

100, 102, 106, 110, 112, 115, 115, 117, 123, 125

Here, since the number of data-points is even therefore neither the fifth term nor the sixth term is right in the middle. The middle term of data i.e. the median will lie between fifth and sixth terms. So, the median of data in this situation is the average of the two middle terms. In this example-

Fifth term = 112 *cm*

Sixth term = 115 *cm*

$$\text{Median} = \frac{\textit{Fifth term} + \textit{Sixth term}}{2}$$

$$= \frac{112+115}{2} = 113.5 \ cm$$

Median of this data is 113.5 *cm*

This means that when number of data-points is even then we can understand median as follows:

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} \textit{term} + \left(\frac{n}{2}+1\right)^{th} \textit{term}}{2}$$

So far we have calculated median of given observations. Now we will use median to find missing observations.

Example-8. Median of the data series 7, 10, 12, *p*, *q*, 27, 31 arranged in ascending order is 17. If one more data-point 40 is included then median becomes 18. Find the value of *p* and *q*.

Solution : You know that median is always the middle term of a data series.

In the series 7, 10, 12, *p*, *q*, 27, 31 the fourth term lies in the middle and its value will be the median. We are given that the median is 17 and since the fourth term is *p*, so $p = 17$.

If one more observation 40 is included in the series then we get 7, 10, 12, *p*, *q*, 27, 31, 40. Now number of data-points becomes even so

$$\text{New Median} = \frac{\left(\frac{n}{2}\right)^{th} term + \left(\frac{n}{2}+1\right)^{th} term}{2}$$

$$18 = \frac{p+q}{2}$$

$$18 = \frac{17+q}{2}$$

$$36 = 17 + q$$

$$19 = q$$

So, the values of p and q are 17 and 19 respectively.

Mode

You read about average and median. One more method of to derive conclusions from any data is finding its 'mode'. Mode is the data point that occurs the maximum number of times in a series. For example, in a Class-X examination, marks obtained by 20 students are:-

40, 25, 40, 35, 36, 45, 45, 40, 35, 39, 41, 42, 40, 25, 40, 42, 35, 38, 40

After looking at the data we find that maximum number of students (6) got 40 marks so mode is 6. Let consider one more example:

Example-9. A shopkeeper sells shoes of different sizes (6, 7, 8, 9, 10) in his shop. Sales data in three months is as follows

Shoe size	6	7	8	9	10
Number of shoes sold	18	24	41	19	9

The shopkeeper saw that many shoes had been sold in three months. Now shopkeeper wanted to restock his shoes. Can he use mean and median to decide which shoe-size should be ordered from the shoe-company at the earliest to complete his stock? He can't find this from mean and median. He has to reorder the shoes in the size that is sold more frequently.

On considering above records, the shopkeeper decided to order shoes in size 8. For the time being, he decides not to buy shoes in other sizes. You can see that the maximum demand of shoes is in size 8 from the number of shoes sold.

Therefore, here mode is 8.

Try These

Find the mode of the following data:-

1. 25, 9, 69, 34, 70, 36, 90, 70, 56, 70, 71

2. 56, 39, 94, 36, 39, 15, 39, 40

Exercise - 1

1. To find the answers to the following questions will you use arithmetic mean or median? In which question(s), we can't use either of the two?
   - (i) Which is the most popular newspaper in the state?
   - (ii) What was the average rainfall in a month?
   - (iii) 100 students participated in an examination. Who are the top-performing 50 students among them depending on marks obtained?
   - (iv) What was the average price of petrol in the month of January?
   - (v) Which player has taken highest number of wickets in international cricket matches?
   - (vi) How will you decide the number of rotis to be cooked for 20 guests invited to a party?
   - (vii) In which month does maximum rainfall occur?

2. Data of rainfall (*mm*) in 10 months is as follows:-

   243.50, 266.00, 347.70, 240.00, 325.20,
   264.80, 356.30, 211.60, 246.90, 282.70

   Find the average rainfall from the data.

3. Name the first 10 even numbers. Find their average.

4. Find the average price of rice in five different states from the given data-

City	A	B	C	D	E
Price (In Rupees)	25	28	30	31	32

5. The table below gives data for high jump in international games (Olympics). Find the mean, mode and median of the data.

Year	1960	1964	1968	1972	1976	1980	1984	1988	1992	1996	2000	2004
Height (in meter)	1.85	1.90	1.82	1.92	1.93	1.97	2.02	2.03	2.02	2.05	2.01	2.06

6. Weight (in kilograms) of eight students is as follows-

30, 32, 33, 38, 37, 41, 35, 40

Find the average weight of the students.

7. Number of students enrolled in a school over five consecutive years is as follows-

1150, 1250, 1360, 1275, 1310

What was the average number of students in the school in this five-year period?

Limitations of Arithmetic Mean, Median and Mode

To interpret data one representative value is the arithmetic mean. We saw that mean gives us a lot of information about the data, but many aspects remain unclear and mistakes can happen if we use mean blindly in all situations. For example, we cannot decide the height of doors in a house by taking the average height of all family members. Nor can we decide it by finding out the most common height of the family members.

Besides the limitations of using mean, we saw that median and mode are also unable to give answers to all our questions. These are helpful in understanding data but we should use these very carefully.

Measures of Central Tendency in Grouped Data

In most cases the number of observations is so large that to interpret and read it correctly we first have to reduce its complexity by making groups (by classifying it). After converting it into group data we can find mean, median and mode to interpret it.

In example-13, grouped data is given by taking class interval of 10. It should be remembered that while determining frequency of class interval, any data which is equal to upper limit of a class interval will be placed in the next class interval. For example, the year in which production of rice was 50 lacs ton is not placed in the interval 40-50 but in the class interval 50-60.

We saw that to calculate the mean of ungrouped data, we first sum all data-points. But what we will do for grouped data? Which value should we take from a group, which number should we choose? For class interval 40-50, should we take 40 or 50 or any other value?

So, we need values which will represent the different groups. We assume that frequencies of all class intervals are centred around the mid-point and mid-point of each class interval is representative of that class interval. We can also call these mid-points class marks.

Example-10. Data of weight of young and old groups of children of a senior secondary school is given below. Find the average weight per child.

Weight (In *kg*)	30—40	40—50	50—60	60—70	70—80
Number of students	11	29	6	3	1

Solution: First we find mid-point. To find mid-point we have to use class-limit. Mid-point is the average of upper limit and lower limit of class. Mid-point of the class (30-40) is 35, i.e.

$$\text{Mid-point} = \frac{\textit{Lower Limit} + \textit{Upper Limit}}{2} = \frac{30+40}{2} = 35$$

We denote mid-point by x_i. First mid-point x_1 is 35.

In the same manner we can also find mid-points of other classes and they are 45, 55, 65 and 75 respectively. Now, we multiply each mid-point with corresponding frequency and use it to calculate mean. The new table will be as follows:

Weight (*kg*)	**Number of students** (f_i)	**Mid-point** (x_i)	$(f_i x_i)$
30—40	11	35	385
40—50	29	45	1305
50—60	6	55	.....
60—70	3	65	.....
70—80	1	75	.....
योग	**50**		**2290**

Complete the table.

So, we find that the sum of $f_i x_i$ in the above table i.e. $\Sigma f_i x_i = 2290$.

So, the mean $\overline{X}$ of the given data will be:

$$\overline{X} = \frac{\sum f_i x_i}{\sum f_i} = \frac{2290}{50} = 45.8 \ kg$$

Thus, average weight per child is 45.8 *kg*.

Mode of Grouped Data

In the example given above, we found weight of children on an average. If we want to know how much most children weigh then we will have to find the mode.

You know that mode is that data value which is seen most frequently. In grouped data, we first find the group containing the mode. In given data, the frequency of class (40-50) is greatest so it is the mode class. This means that mode is present in this class interval. In such situations, we can find mode by putting the values in the following formula.

The more the difference between f_1 and f_o greater is the mode than *l*. Similarly, lesser the difference between f_1 and f_2, more is the distance between *l* and mode and closer is mode to *l+h*. If we think about what can be the maximum value of mode we will find that the maximum value can be the sum of *l* and *h*, which is the difference between f_2 and f_1 or between f_1 and f_0. That is, the mode will lie between *l* and *l+h*.

The formula for finding mode is

$$\text{Mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h$$

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l = lower limit of mode class

f_0 = frequency of the class preceding the modal class.

f_1 = frequency of the modal class.

f_2 = frequency of the class succeeding the modal class.

h = size of the class interval.

Example-11. In the table of example 10, the modal class = 40–50, the lower limit of modal class (l) = 40

frequency of the modal class (f_1) = 29

frequency of the class immediately preceding the modal class (f_0) = 11

frequency of the class immediately succeeding the modal class (f_2) = 6

Size of the class interval (h) = 10

By putting these values in the formula

$$\text{Mode} = 40 + \left[\frac{29-11}{2(29)-11-6}\right] \times 10$$

$$= 40 + \left[\frac{18}{58-17}\right] \times 10 = 40 + \frac{18}{41} \times 10$$

$$= 44.39\ kg$$

This mode is near to $l+d$ because f_0 is big and f_2 is small.

In this way we can find mode of grouped data which is close to observations.

Median of Grouped Data

Example-12. Height of girls in X class of a school is as follow:-

Height (*cm*)	135—140	140—145	145—150	150—155	155—160
Number of Girls	1	2	11	9	7

Find the median of these data.

Solution: To find the median from given data we have to find cumulative frequency from frequency (you have learnt how to find cumulative frequency in class IX).

Height	**Number of girls (Cumulative frequency)**
less than 140	1
less than 145	1 + 2 = 3
less than 150	3 + 11 = 14
less than 155	14 + 9 = 23
less than 160	23 + 7 = 30

This cumulative frequency distribution is of the type 'less than' where 140, 145, 150, 155, 160 are the upper limits of the class.

We know that the mid-point observation of the given data will lie is in some class interval. How do we find that class interval in which middle data point is present?

Height	**Number of girls (*f*)**	**Cumulative frequency (*cf*)**
135—140	1	1
140—145	2	3
145—150	11	14
150—155	9	23
155—160	7	30

To find median class we find cumulative frequency of each class and $\frac{n}{2}$ Now we find that class whose cumulative frequency is greater than or very near $\frac{n}{2}$ Here, this is $n = 30$ i.e. $\frac{n}{2} = 15$. Since, 150-155 is that class whose cumulative frequency, 23 is more than 15 therefore median class will be 150-155.

Thus, median class is 150-155. After finding median class we can find median by using the following formula-

$$\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$$

Where, l = lower limit of median class

n = number of observation

cf = cumulative frequency of class preceding the median class

f = frequency of median class

h = class size (assuming class sizes to be equal)

Now, $\frac{n}{2}=15$, $l = 150$, $cf = 14$, $f = 9$, $h = 9$

$$\text{Median} = 150+\left(\frac{15-14}{9}\right)\times 5$$

$$=150+\frac{5}{9}$$

$$=150.55 \quad cm$$

So, the height of about half of the girls is less than 150.55*cm* and the height of remaining half is more than or equal to 150.55*cm*.

In the same way we can also arrange data in the form of 'more than'. This is shown in the table. We can interpret many things from this table, for example, the height of 16 girls is more than 150*cm* etc.

Height	Number of girls
More than or equal to 135	30
More than or equal to 140	29
More than or equal to 145	27
More than or equal to 150	16
More than or equal to 155	7

What conclusion can you draw from this table? Discuss and find 3 conclusions.

Data Trends: Extrapolation and interpolation

We find that after organizing and interpreting data we get many types of information but not all. One more question that can be asked is whether we can say something about data beyond the data given in each class interval? Imagine that we have kept records of total rainfall in a city for certain number of years. But the data of rainfall of some years could not be collected and hence, is missing. So, can we figure out what these missing data are? Also, can we predict how much it will rain in the coming years on the basis of this data?

To answer these two questions, we consider the pattern shown by the data. Is there a pattern to the change of data? Can we see certain trends in the data? We will look at the few examples to understand this better and to also see where we can apply extrapolation and where we can't?

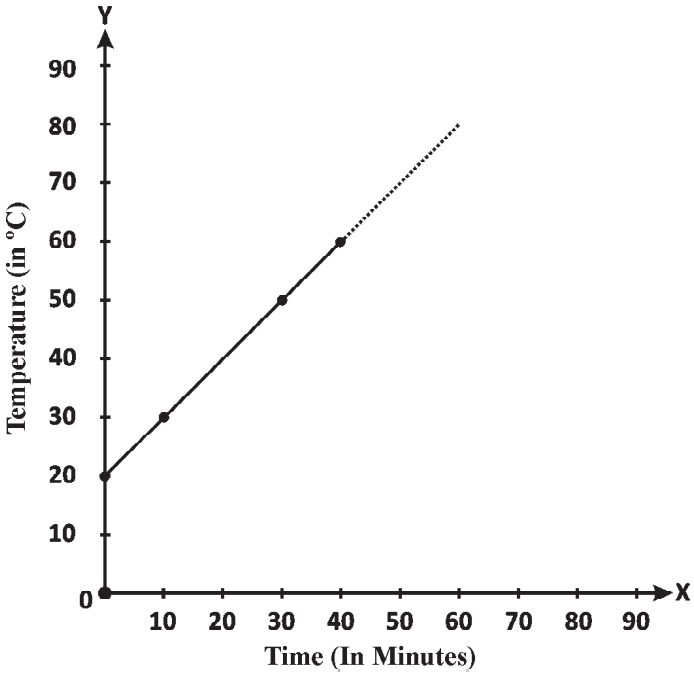
Suppose you heat a liquid for 40 minutes and record its temperature at four different times as shown in the table:-

Time (in minutes)	0	10	30	40
Temperature (in degree Celsius)	20	30	50	60

If we represent this data on graph we get some points. Draw a curve that joins all the points. We can say that in the beginning temperature was 20°C at 0 minute, 30°C after 10 minutes, 50°C after 30 minutes, and 60°C after 40 minutes but can we say something about the temperature after 20 minutes or 60 minutes by looking only at the data. Obviously not! Temperature is given only on minutes 0, 10, 30 and 40. 20 minutes come in the 10-30 interval. With the help of the graph we can determine the corresponding temperature at 20 minutes and that is 40°C (Graph-2).

Here, in the given data there are infinite points which lie within the given data range for which we can find the corresponding temperature on the graph. This is known as interpolation.

Notice that here range of data is between 0-40 (minutes). The time point 50 minutes falls after this range. To know the



temperature at 50 minutes we will extend the graph lines in its direction. On extending the line we get the temperature 70°C at 50 minutes. Other corresponding temperatures can also be shown on extending the line in the same manner. This method in which we can find values beyond the data range by following the known trend is known as extrapolation. It is assumed that trend in change of data is uniform and there is no any unacceptable fluctuation in data within and beyond the range of data.

Limitations of Interpolation and extrapolation

In the above example, on the basis of graph we were able to find the temperature at 20 minutes which is between the data range 0 to 40 and on 50 minutes which was outside the range. Can you tell about temperature of liquid when it is boiled for 90 minutes? From the graph, we can see that temperature of the liquid increases by 10°C every ten minutes and on extrapolation the temperature will be 110°C at 90 minutes. Is it possible for water being boiled in an open vessel to reach this temperature? It is obvious that the trend of this graph will change slowly after some time and we can't extrapolate in an unlimited manner in this data.

Second question is whether we can do interpolation and extrapolation with any kind of data. Can we do interpolation and extrapolation in the following data? Let us consider-

Data of maximum high jump in Olympics is as follows-

Year	1960	1964	1972	1976	1980
Height (in meter)	1-85	1-90	1-92	1-93	1-97

On the basis of this data can you tell us what the record for high jump was in the year 1956?

It is not possible because there is no any trend in this data. This is data of the height of jumps which is recorded during the competition. In the same way if there is no trend in the population data, then we can't predict population for the following years or what the population was between the given years. It means there are limitations to interpolation and extrapolation of data due to trends of data and we cannot carry out interpolation and extrapolation on all types of data.

Think and Discuss

Result of Class-X exams are given in the table below:-

Years	2001	2002	2003	2005
Result	88%	80.5%	66%	55%

Can you figure out the results in the years 2004 and 2006?

Getting Data from Graphs

Try These

Collect height (cm) and age data of 40 boys and girls in different classes in your school and draw the graph between their ages and age-wise average height (be careful while taking data that each age group contains at least 3 to 5 children).

Answer the following on the basis of your graph-

1. Average height of 15 year old girls?

2. Average height of 10 year old girls?

3. What is the increase in the average height of the girls from the age of 13 years to 15 years?

4. Can you find the average height of 14 year old girls? How?

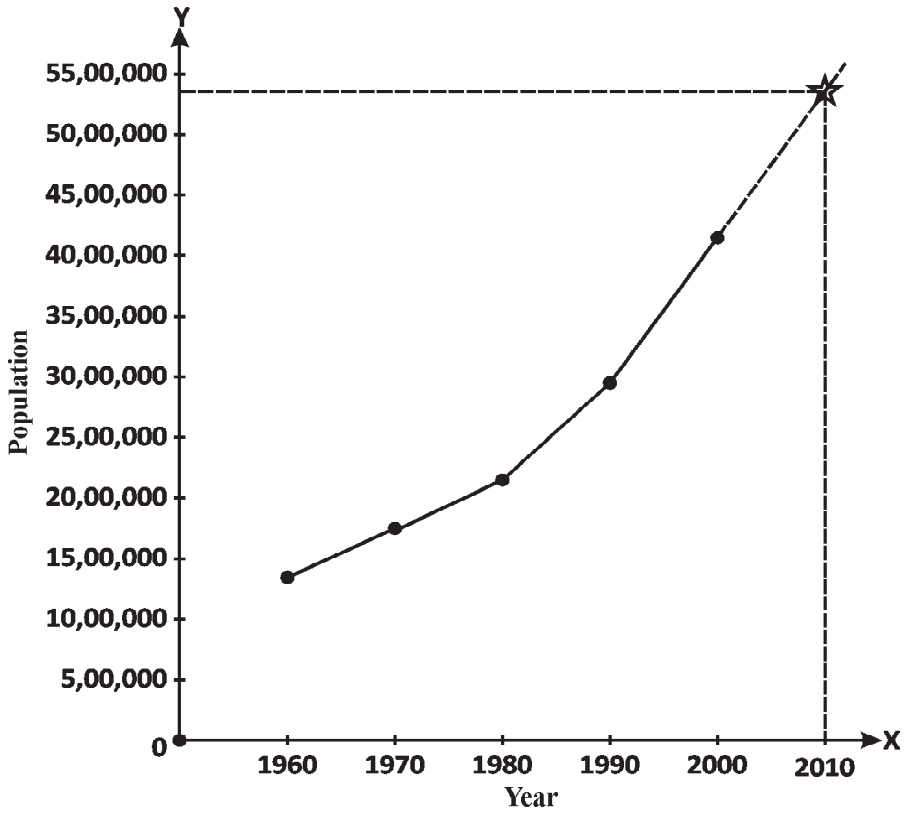
5. What is the average height of 16 years old girls?

   How can we find the answers of questions 4 and 5?

Hint: Show 14 years on the X-axis. Draw a perpendicular from this point on the X-axis to a point on the graph. Draw a straight line from this point on the graph to the Y axis. The value on the y-axis where the line meets it will show the average height of 14 year old girls. The same method can be used to estimate the average height of girls or children of any age group from the graph.

6. Can you use the same graph to estimate the average height of 6 year old girls or 20 year old girls? Discuss and write your answer.

Example-13. Consider the following graph which is related to population and give the answers of the questions on the basis of this graph.



(i) What was the population in the year 1980?

(ii) What is the growth in population from 1960 to 2000?

(iii) What was the population in the year 1975?

(iv) What was the population in the year 1995? Find out.

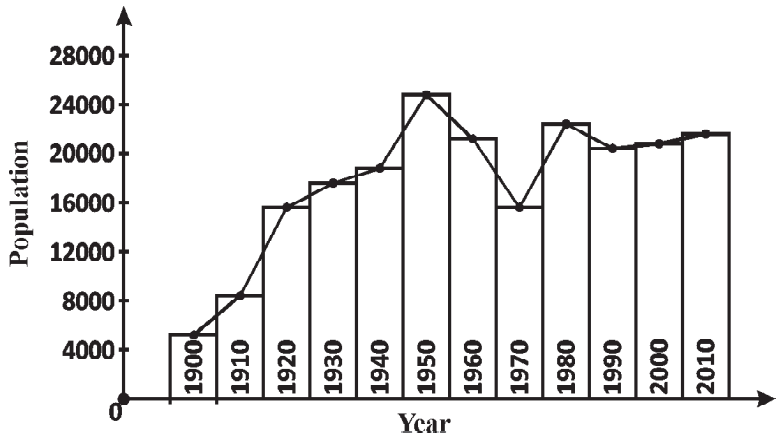
(v) Does on the basis of given data, can we estimate the population in the year 2010?

We can get the answers of questions (i) and (ii) directly from the graph. But the data related to questions (iii), (iv) and (v) will have to be extracted from the graph. If we consider the data in the graph it is continuously increasing. Besides showing years 1975, 1995 and 2010 on X-axis, we will also have to extend the scale from 4,500,000 to 5,500,000 on the Y-axis. Now if we join the points corresponding to years 1975, 1995 and 2010 on the graph to corresponding point on Y-axis, then we get estimated population data for the year (see graph). Thus, we get estimated data on the basis of recorded data.

Estimated data helps us plan for the future in the present. For example, population control can be planned in the present before it touches the estimated mark in reality.

***Note*:-** It is supposed on the basis of estimated data for the year 2010 that the trend of increase in population will remain the same. It is not necessary that this will happen because many methods are being used to control population even now. How these efforts will affect the trend of data cannot be determined, we can only estimate on the basis of extrapolation.

Example-14. The population of a country over different years is shown in the graph. Analyze the graph and answer the following questions.



(i) In which year was the population maximum?

(ii) What is the minimum population?

(iii) In which years did population show an increase?

(iv) In which years did the population fall?

(v) Was there a constant growth or fall in population in the initial five years?

Let us think more about this data.

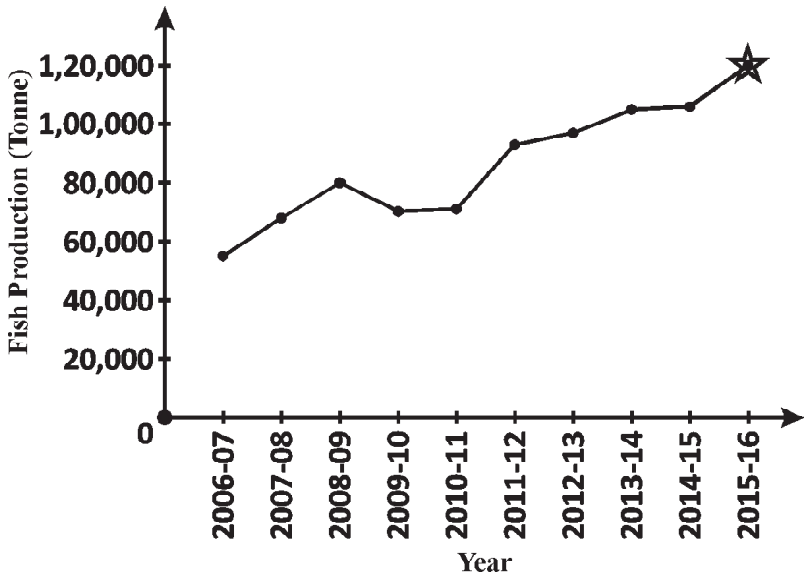
If we consider data of first fifty years from 1900 to 1950 we see that population is continuously increasing. If we look at the population figures after this then population decreased from 1950 to 1970 and after that again recorded growth. The data from 1990 to 2010 also shows increasing population.

We find that the graph of population growth changed directions continuously. In different years the population sometimes increased and sometimes decreased and the direction of rate of growth changed – this is known as data trend. Data trends help us predict the value of data in the future.

Let us look at another example.

Example-15. The following graph shows figures related to production of fishes in a state.

Analyze the graph and answer the following questions:-



(i) How many ton of fish were produced in the year 2011-12?

(ii) In which year was the production of fish maximum?

(iii) Do we see a continuous increase in fish production?

In 2007, the state initiated a scheme called 'friend of fish' in which the ichthyology (fish) department tried to motivate the local community towards fish farming. Now, several thousand people from the community are involved with the department. They help the department identify suitable locations for cultivating fish and also play a role in collecting and maintaining important data about fish production.

Because of active involvement of the community, the state has seen a constant increase in production. If we extrapolate on the basis of previous figures, it is estimated that fish production will touch 1,20,000 ton mark in 2015-16.

Exercise - 2

1. Find the mean of the following data:-

Number of female teacher (in %)	15–25	25–35	35–45	45–55	55–65	65–75	75–85
No. of states	6	11	7	4	4	2	1

On the basis of this data draw 5 conclusions about female teachers.

2. Wickets taken by some bowlers in international cricket matches are given in the table. Find the mode of the data.

Number of wicket	0–50	50–100	100–150	150–200	200–250	250–300
Number of bowlers	4	5	16	12	3	2

Write 5 conclusions about this data.

3. Literacy rates (in percentage) of 35 cities are given in the following table. Find the mean of the data.

Literacy Rate (in %)	45–55	55–65	65–75	75–85	85–95
Number of Cities	3	10	11	8	3

Write 3 conclusions about this data.

4. The following table shows the ages of the patients admitted in a hospital during a year. Find the mean.

Age (in years)	5–15	15–25	25–35	35–45	45–55	55–65
Number of Patients	6	11	21	23	14	5

Write 3 conclusions about this data.

5. The following table shows marks obtained by the students in an examination-

Marks obtained	0-10	10-20	20-30	30-40	40-50	50-60
Number of students	1	12	24	32	10	5

Find the median of marks. Write 3 conclusions about the data.

What We Have Learnt

1. Bar graphs and tables are helpful in understanding data.
2. Observations are data-points under consideration in a particular situation.
3. Average is a number which is characteristic of the entire data.
4. Average lies between the maximum and the minimum value of the given data.
5. Average of data is known as arithmetic mean in statistics.
6. Mean, median and mode are representative values of data.
7. Median is the middle value of data organized in ascending or descending order.
8. Mode is the value that occurs maximum number of times in given data.
9. If there is lots of variation in data then mean of that data can be misleading.
10. To find median of series having individual observations we have to arrange data in ascending or descending order.
11. The difference between highest and lowest values (limits) of data is said to be its range. It shows the extent of variation in the data points.
12. The formula to find arithmetic mean in individual series is as follows:

$$\text{Arithmetic Mean} = \frac{Sum\ of\ data}{Number\ of\ data}$$

13. To find arithmetic mean of discrete and grouped series, the following formula is used:

$$\bar{X} = \frac{\sum f_i x_i}{\sum f_i}$$

14. To find median of individual series, if number of terms is odd then we use the following formula-

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} term$$

And we use the following formula if number of data is eve

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} term + \left(\frac{n}{2}+1\right)^{th} term}{2}$$

16. To find median of grouped data we use the following formula-

$$\text{Median} = \ell + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$$

17. To find mode of grouped data we use the following formula-

$$\text{Mode} = \ell + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

18. With the help of a graph we can find data-points within the data range which are actually given. This is known as interpolation but it is not possible all data types.

19. In graphs, when we use the direction of increase or decrease of the curve to determine a value beyond the given data range, this is known as extrapolation. Extrapolation can be done only if the data shows a clear trend.

Answer Key - 1

1. (i) Mode (ii) Arithmetic Mean (iii) Median
(iv) Arithmetic mean (v) Mode (vi) Arithmetic mean
(vii) None of these

2. 278.47 *mm* 3. 11 4. Rupees 29.2

5. Arithmetic mean = 1.965 meter, median = 1.99 meter, mode = 2.02 meter

6. 35.75 kilogram

Answer Key - 2

1. 39.71% 2. 136.66 3. 69.42%

4. Mean = 35.37 5. 31.56